

Question 1: (15 Marks) (START A NEW PAGE)

- (a) Given $z = 3 + 2i$ and $w = 1 + 3i$. Express the following in the form $z = a + ib$ where a and b are real numbers

(i) $z + \bar{w}$ 2

(ii) z^2 2

(iii) $\frac{z}{w}$ 3

- (b) (i) Express $1 + i\sqrt{3}$ in mod-arg form. 2

(ii) Simplify $(1 + i\sqrt{3})^9$. 2

- (c) (i) Show that $z = i$ is a root of the quadratic equation $(2 - i)z^2 - (1 + i)z + 1 = 0$. 1

(ii) Find the other root of the above equation in the form $z = c + id$ where c and d are real numbers. 3

Question 2: (15 Marks) (START A NEW PAGE)

- (a) Find real values for x and y if $(x + iy)(3 + i) = 5i$ 2

- (b) Make neat sketches of the following loci on separate diagrams.

(i) $|z| = |z - 4 - 2i|$. 2

(ii) $\arg(z - i) = \frac{\pi}{4}$. 2

(iii) $|z - 3 + 4i| \leq 5$. 3

- (c) Let $z = x + iy$ be any non-zero complex number.

(i) Express $z + \frac{1}{z}$ in the form $p + iq$. 2

(ii) Given that $z + \frac{1}{z}$ is real, show that $y = 0$ or $x^2 + y^2 = 1$. 2

(iii) If $y = 0$, show that $\left|z + \frac{1}{z}\right| \geq 2$. 2

Question 3: (15 Marks) (START A NEW PAGE)

- (a) (i) Find both square roots of $z = 5 - 12i$, writing your answers in the form $z = x + iy$ where x and y are real numbers. 2
- (ii) Hence solve $z^2 + z - 1 + 3i = 0$ writing your answers in the form $z = p + iq$ where p and q are real numbers. 3
- (b) (i) Express $1 + i$ in mod-arg form 2
- (ii) Find the smallest positive value of m if $(1 + i)^m = (1 - i)^m$. 3
- (c) z_1 , z_2 and z_3 are the three cube roots of a complex number z and $|z| = 1$.
- (i) Assuming that $\arg(z_1) = \theta$ where $0 \leq \theta \leq \frac{\pi}{2}$, draw a diagram clearly showing the positions of the complex numbers z_1 , z_2 and z_3 . 2
- (ii) Evaluate $|z_1 - 1|^2 + |z_2 - 1|^2 + |z_3 - 1|^2$. 3

Question 4: (15 Marks) (START A NEW PAGE)

- (a) (i) Given $z_1 = \cos \alpha + i \sin \alpha$, show that $1 + z_1 = 2 \cos \frac{\alpha}{2} (\cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2})$. 2
- (ii) If $z_2 = \cos \beta + i \sin \beta$ find, in terms of α and β , the modulus and argument of the complex number $\frac{z_2 + 1}{z_1 + 1}$. 2
- (iii) For two complex numbers $z_1 (\neq -1)$ and $z_2 (\neq -1)$, $|z_1| = |z_2| = 1$ and $\frac{z_2 + 1}{z_1 + 1} = i$. Find z_1 and z_2 in the form $z = x + iy$ where x and y are real numbers. 4
- (b) (i) Find the nine roots of $z^9 - 1 = 0$, leaving your answers in mod-arg form. 2
- (ii) If $P(z) = z^6 + z^3 + 1$, explain why the roots of $P(z) = 0$ are also roots of $z^9 - 1 = 0$. 1
- (iii) Show the position of the roots of $P(z) = 0$ on an Argand Diagram. 2
- (iv) Express $P(z)$ in the form $P(z) = (z^2 - az + 1)(z^2 - bz + 1)(z^2 - cz + 1)$ where a , b and c are real numbers. 2

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Year 11 Term 4 Assessment – Extension II – 2002

Question 1:

$$\begin{aligned} 1(a) \quad (i) \quad z + \bar{w} &= (3 + 2i) + (1 - 3i) \\ &= 4 - i \end{aligned}$$

$$\begin{aligned} (ii) \quad z^2 &= (3 + 2i)^2 \\ &= 9 + 12i + 4i^2 \\ &= 5 + 12i \end{aligned}$$

$$\begin{aligned} (iii) \quad \frac{z}{w} &= \frac{3 + 2i}{1 + 3i} \times \frac{1 - 3i}{1 - 3i} \\ &= \frac{3 - 9i + 2i - 6i^2}{1 + 9} \\ &= \frac{9 - 7i}{10} \\ &= \frac{9}{10} - \frac{7}{10}i \end{aligned}$$

$$\begin{aligned} 1(b) \quad (i) \quad \tan \theta &= \sqrt{3} \\ \arg(z) = \theta &= \frac{\pi}{3} \\ |z| &= \sqrt{1^2 + (\sqrt{3})^2} \\ &= 2 \\ 1 + i\sqrt{3} &= 2 \operatorname{cis} \frac{\pi}{3} \end{aligned}$$

$$\begin{aligned} (ii) \quad (1 + i\sqrt{3})^9 &= \left(2 \operatorname{cis} \frac{\pi}{3} \right)^9 \\ &= 2^9 \operatorname{cis} 3\pi \\ &= 512(\cos 3\pi + i \sin 3\pi) \\ &= -512 \end{aligned}$$

$$\begin{aligned}
 1(c) \quad (i) \quad P(i) &= (2-i)(i)^2 - (1+i)(i) + 1 \\
 &= (2-i)(-1) - (1+i)(i) + 1 \\
 &= -2 + i - i - i^2 + 1 \\
 &= -2 + i - i + 1 + 1 \\
 &= 0
 \end{aligned}$$

$\therefore z = i$ is a root

(ii) Let other root be $z = \alpha$

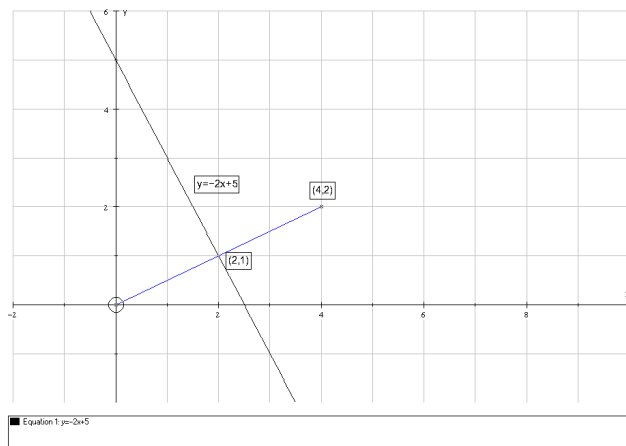
$$\alpha + i = \frac{1+i}{2-i} \quad \left(\text{sum of roots} = -\frac{b}{a} \right)$$

$$\begin{aligned}
 \alpha &= \frac{1+i}{2-i} - i \\
 &= \frac{1+i-2i+i^2}{2-i} \\
 &= \frac{-i}{2-i} \times \frac{2+i}{2+i} \\
 &= \frac{-2i-i^2}{5} \\
 &= \frac{1}{5} - \frac{2}{5}i
 \end{aligned}$$

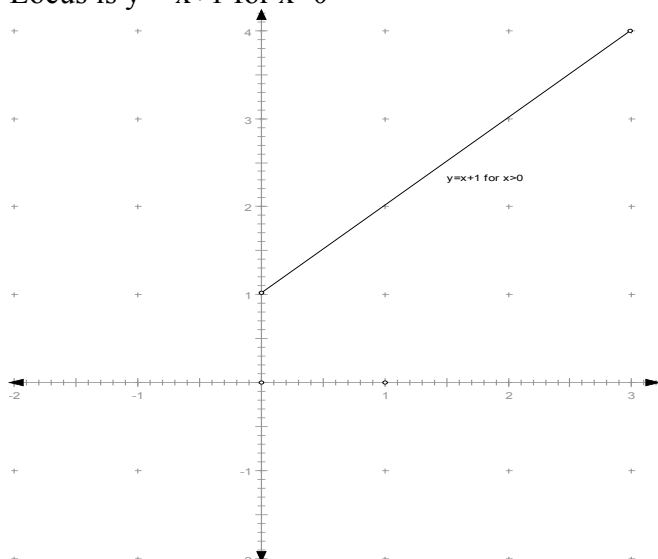
Question 2:

2(a) $(x + iy)(3 + i) = 5i$
 $3x + ix + 3iy + i^2 y = 5i$
 $(3x - y) + (x + 3y)i = 5i$
 $3x - y = 0 \dots\dots\dots(1)$
 $x + 3y = 5 \dots\dots\dots(2)$
from (1) $y = 3x$
sub into (2) $x + 9x = 5$
 $\therefore x = \frac{1}{2}$ and $y = \frac{3}{2}$

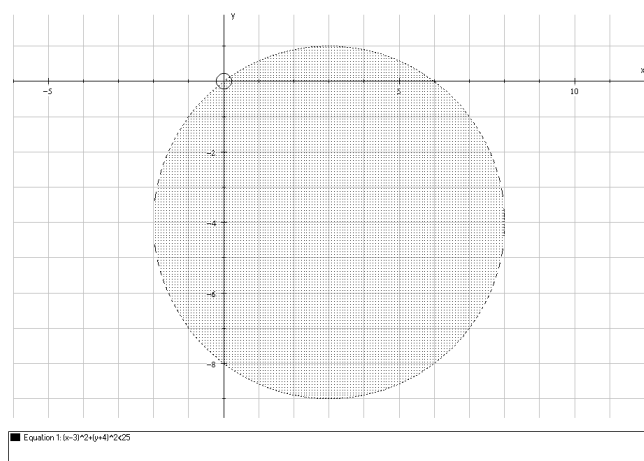
2(b) (i) Locus is $y = -2x + 5$



(ii) Locus is $y = x + 1$ for $x > 0$



(iii) Locus is interior of circle center $(3, -4)$ and radius $= 5$



$$\begin{aligned}
2(c) \quad (i) \quad z + \frac{1}{z} &= (x + iy) + \frac{1}{x + iy} \\
&= x + iy + \frac{x - iy}{x^2 + y^2} \\
&= \frac{x(x^2 + y^2) + iy(x^2 + y^2) + x - iy}{x^2 + y^2} \\
&= \frac{x(x^2 + y^2 + 1)}{x^2 + y^2} + \frac{y(x^2 + y^2 - 1)}{x^2 + y^2}i
\end{aligned}$$

$$(ii) \quad \text{If } z + \frac{1}{z} \text{ is real then } \text{Im}(z) = 0$$

$$\therefore y(x^2 + y^2 - 1) = 0$$

$$\therefore y = 0 \text{ or } x^2 + y^2 = 1$$

$$\begin{aligned}
(iii) \quad \text{If } y = 0 \text{ then } z + \frac{1}{z} &= \frac{x(x^2 + 1)}{x^2} \\
&= x + \frac{1}{x}
\end{aligned}$$

$$\begin{aligned}
\therefore \left| z + \frac{1}{z} \right| &= \left| x + \frac{1}{x} \right| \\
&= \left| x + \frac{1}{x} - 2 + 2 \right| \\
&= \left| \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 + 2 \right| \text{ for } x > 0 \\
&\geq 2 \quad \text{since } \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 \geq 0
\end{aligned}$$

by symmetry, result is also true for $x < 0$

Question 3:

3(a) (i) Let $z = a + ib$

$$(a + ib)^2 = 5 - 12i$$

$$\therefore a^2 - b^2 + 2iab = 5 - 12i$$

equating terms

$$a^2 - b^2 = 5 \dots\dots\dots (1)$$

$$ab = -6 \dots\dots\dots (2)$$

by inspection $a = 3, b = -2$ or $a = -3, b = 2$

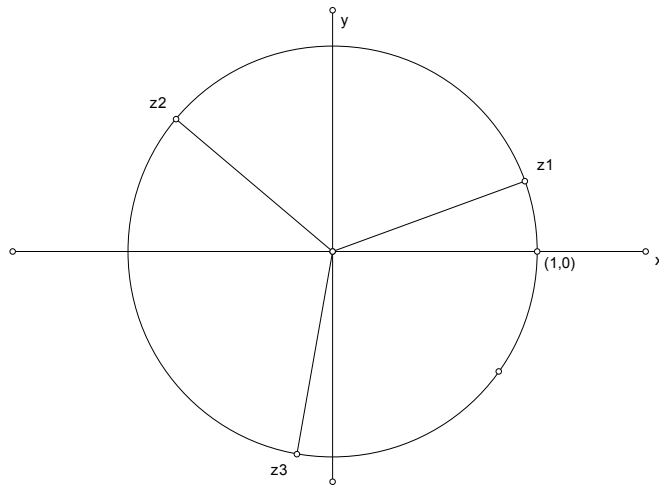
$$\therefore z = 3 - 2i \text{ or } z = -3 + 2i$$

$$\begin{aligned} \text{(ii)} \quad z &= \frac{-1 \pm \sqrt{1 - 4(-1 + 3i)}}{2} \\ &= \frac{-1 \pm \sqrt{5 - 12i}}{2} \\ &= \frac{-1 + (3 - 2i)}{2} \text{ or } \frac{-1 - (3 - 2i)}{2} \\ z &= 1 - i \text{ or } -2 + i \end{aligned}$$

$$3(b) \quad (i) \quad 1 + i = \sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right)$$

$$\begin{aligned} \text{(ii)} \quad 1 - i &= \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) \\ \therefore \left\{ \sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right) \right\}^m &= \left\{ \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) \right\}^m \\ \left\{ \sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right) \right\}^m \div \left\{ \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) \right\}^m &= 1 \\ \left\{ \operatorname{cis}\left(\frac{m\pi}{4}\right) \div \operatorname{cis}\left(-\frac{m\pi}{4}\right) \right\} &= 1 \\ \operatorname{cis}\left(\frac{m\pi}{2}\right) &= 1 \\ \frac{m\pi}{2} &= 0, 2\pi, 4\pi, 6\pi, \dots\dots\dots \\ m &= 0, 4, 8, 12, \dots\dots\dots \\ \therefore \text{smallest positive value is } m &= 4 \end{aligned}$$

3(c) (i)



$$\begin{aligned}
 \text{(ii)} \quad & |z_1 - 1|^2 + |z_2 - 1|^2 + |z_3 - 1|^2 \\
 &= (z_1 - 1)(\bar{z}_1 - 1) + (z_2 - 1)(\bar{z}_2 - 1) + (z_3 - 1)(\bar{z}_3 - 1) \\
 &= z_1 \bar{z}_1 - (z_1 + \bar{z}_1) + 1 + z_2 \bar{z}_2 - (z_2 + \bar{z}_2) + 1 + z_3 \bar{z}_3 - (z_3 + \bar{z}_3) + 1 \\
 &= 6 - (z_1 + z_2 + z_3) - (\bar{z}_1 + \bar{z}_2 + \bar{z}_3) \\
 &= 6 \text{ since } z_1 + z_2 + z_3 \text{ (and the conjugates) is the sum of the roots of } z^3 = 1 \text{ and } \therefore = 0
 \end{aligned}$$

or

$$\begin{aligned}
 & |z_1 - 1|^2 + |z_2 - 1|^2 + |z_3 - 1|^2 \\
 &= [2 + 2 \cos \theta] + [2 + 2 \cos(\frac{2\pi}{3} + \theta)] + [2 + 2 \cos(\frac{2\pi}{3} - \theta)] \\
 &= 6 + 2[\cos \theta + \cos(\frac{2\pi}{3}) \cos \theta - \sin(\frac{2\pi}{3}) \sin \theta + \cos(\frac{2\pi}{3}) \cos \theta + \sin(\frac{2\pi}{3}) \sin \theta] \\
 &= 6 + 2[\cos \theta + 2 \cos(\frac{2\pi}{3}) \cos \theta] \\
 &= 6 + 2[\cos \theta + 2(-\frac{1}{2}) \cos \theta] \\
 &= 6 + 2[\cos \theta - \cos \theta] \\
 &= 6
 \end{aligned}$$

Question 4:

$$\begin{aligned}
 4(a) \quad (i) \quad 1 + z_1 &= 1 + \cos a + i \sin a \\
 &= 2 \cos^2\left(\frac{a}{2}\right) + 2i \sin\left(\frac{a}{2}\right) \cos\left(\frac{a}{2}\right) \\
 &= 2 \cos\left(\frac{a}{2}\right) \left\{ \cos \frac{a}{2} + i \sin \frac{a}{2} \right\}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \frac{z_2 + 1}{z_1 + 1} &= \frac{2 \cos \frac{\beta}{2} \left(\cos \frac{\beta}{2} + i \sin \frac{\beta}{2} \right)}{2 \cos \frac{\alpha}{2} \left(\cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2} \right)} \\
 &= \frac{\cos \frac{\beta}{2}}{\cos \frac{\alpha}{2}} \operatorname{cis}\left(\frac{\beta - \alpha}{2}\right) \\
 \left| \frac{z_2 + 1}{z_1 + 1} \right| &= \frac{\cos \frac{\beta}{2}}{\cos \frac{\alpha}{2}} \quad \text{and} \quad \arg\left(\frac{z_2 + 1}{z_1 + 1}\right) = \frac{\beta - \alpha}{2}
 \end{aligned}$$

$$(iii) \quad |i| = 1 \quad \arg i = \frac{\pi}{2}$$

$$\therefore \frac{\beta - \alpha}{2} = \frac{\pi}{2}$$

$$\beta = \alpha + \pi$$

$$\frac{\cos \frac{\beta}{2}}{\cos \frac{\alpha}{2}} = 1$$

$$\frac{\cos\left(\frac{\alpha}{2} + \frac{\pi}{2}\right)}{\cos \frac{\alpha}{2}} = 1$$

$$\frac{-\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}} = 1$$

$$\tan \frac{\alpha}{2} = -1 \quad \text{for} \quad -\pi < \alpha \leq \pi$$

$$\frac{\alpha}{2} = -\frac{\pi}{4}$$

$$\alpha = -\frac{\pi}{2} \quad \text{and} \quad \beta = \frac{\pi}{2}$$

$$z_1 = \operatorname{cis}\left(-\frac{\pi}{2}\right) = -i \quad \text{and} \quad z_2 = \operatorname{cis}\left(\frac{\pi}{2}\right) = i$$

4(b) (i) $z^9 = 1$

let roots be $z = r \operatorname{cis} \theta$

$$\therefore r^9 \operatorname{cis} 9\theta = \operatorname{cis} 2k\pi, k = 0, 1, 2, 3, 4, 5, 6, 7, 8$$

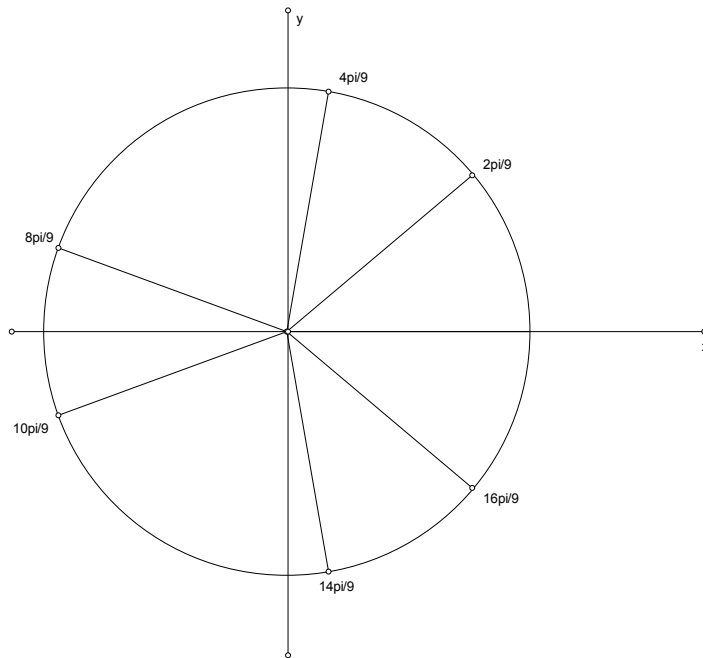
$$r = 1, \quad 9\theta = 2k\pi, k = 0, 1, 2, 3, 4, 5, 6, 7, 8$$

$$z = \operatorname{cis} 0, \operatorname{cis} \frac{2\pi}{9}, \operatorname{cis} \frac{4\pi}{9}, \operatorname{cis} \frac{6\pi}{9}, \operatorname{cis} \frac{8\pi}{9}, \operatorname{cis} \frac{10\pi}{9}, \operatorname{cis} \frac{12\pi}{9}, \operatorname{cis} \frac{14\pi}{9}, \operatorname{cis} \frac{16\pi}{9}$$

(ii) $z^9 - 1 = (z^3)^3 - 1$
 $= (z^3 - 1)((z^3)^2 + (z^3) + 1)$
 $= (z^3 - 1)(z^6 + z^3 + 1)$

since $z^6 + z^3 + 1$ is a factor of $z^9 - 1$, the roots of $z^6 + z^3 + 1$ are also roots of $z^9 - 1$.

(iii)



(iv) The six roots of $z^6 + z^3 + 1$ come in 3 conjugate pairs

$$\text{Let them be } \alpha = \frac{2\pi}{9}, \bar{\alpha} = \frac{16\pi}{9}, \beta = \frac{4\pi}{9}, \bar{\beta} = \frac{14\pi}{9}, \gamma = \frac{8\pi}{9}, \bar{\gamma} = \frac{10\pi}{9}$$

$$\begin{aligned} z^6 + z^3 + 1 &= (z^2 - 2z \cos \alpha + |a|^2)(z^2 - 2z \cos \beta + |\beta|^2)(z^2 - 2z \cos \gamma + |\gamma|^2) \\ &= (z^2 - 2z \cos \frac{2\pi}{9} + 1)(z^2 - 2z \cos \frac{4\pi}{9} + 1)(z^2 - 2z \cos \frac{8\pi}{9} + 1) \end{aligned}$$